

Information Technology

Spring 2026

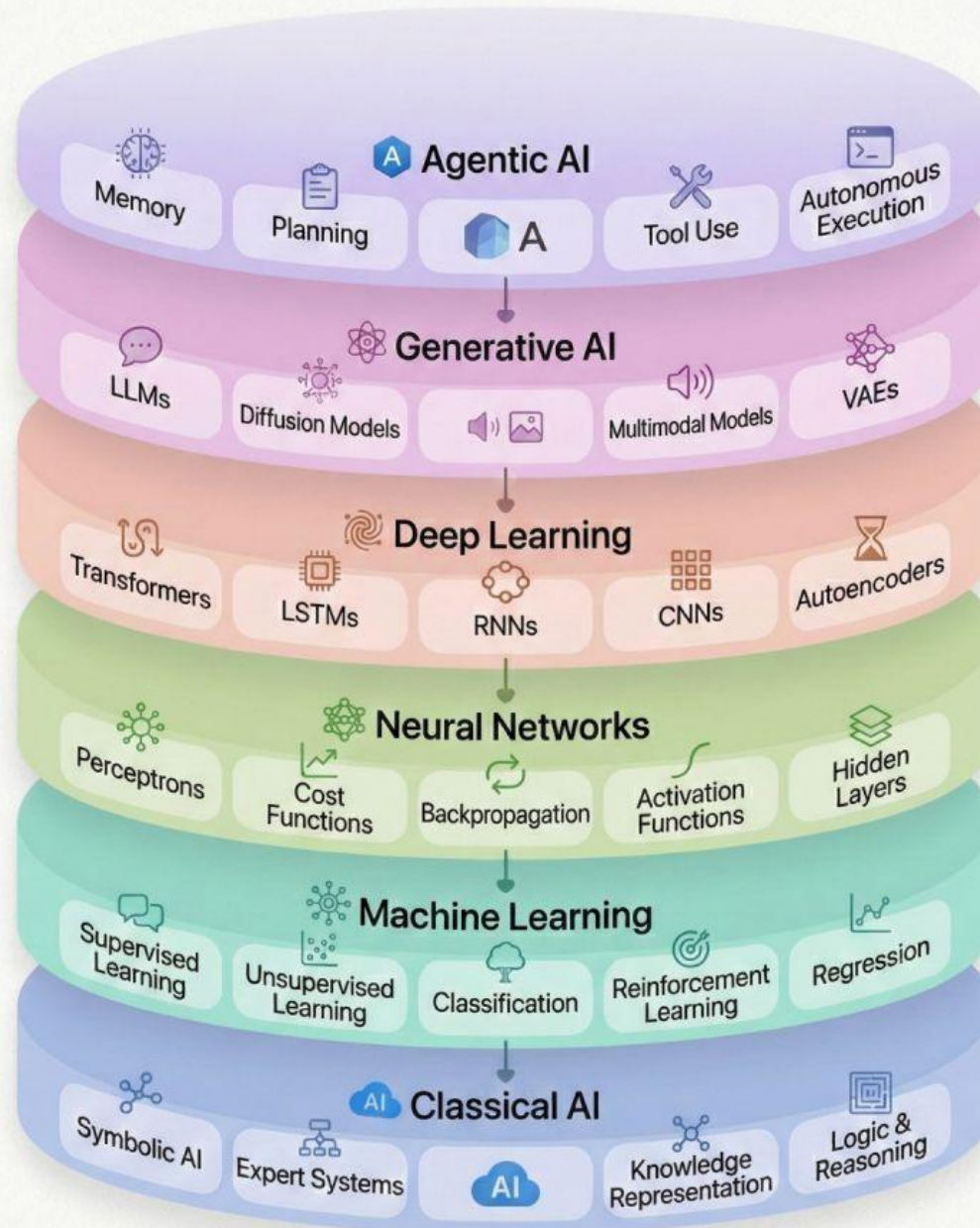
» **Selected Labs in AI**

» **Generative AI**



الجامعة المصرية للتعليم الإلكتروني
THE EGYPTIAN E-LEARNING UNIVERSITY

Layers of AI



Generative AI

- **Generative AI** (short for *generative artificial intelligence*) refers to a type of AI that can **create new content**—such as text, images, audio, video, or code—based on patterns it has learned from existing data.
- **Generative AI** can be thought of as a machine-learning model that is trained to **create new data**, rather than making a prediction about a specific dataset.
- A generative AI system is one that learns to generate more objects that look like the data it was trained on.

Generative AI



Definition	AI models that generate new data or content similar to the data they were trained on.
Main Purpose	To produce original outputs like essays, poems, artwork, music, or realistic images.
Key Examples	- ChatGPT (text) - DALL-E (images) - GitHub Copilot (code) - Jukebox (music)
Technology	Commonly built using models like GPT (text), GANs (images), VAEs, or diffusion models.
Applications	Education, art, marketing, programming, healthcare, gaming, simulations, etc.

How generative AI works

Generative AI operates in three phases:

- **Training**, to create a foundation model that can serve as the basis of multiple gen AI applications.

Tuning, to tailor the foundation model to a specific gen AI application.

Generation, evaluation and retuning, to assess the gen AI application's output and continually improve its quality and accuracy.

Evaluate

Training Phase

- Generative AI begins with a foundation model, a deep learning model that serves as the basis for multiple different types of generative AI applications.
- The most common foundation models today are **Large Language Models (LLMs)**, created for text generation applications.
- There are also foundation models for image generation, video generation, and sound and music generation as well as **multimodal** foundation models that can support several kinds of content generation.

def



Training Phase

- To create a foundation model, practitioners train a deep learning algorithm on huge volumes of raw, unstructured, unlabeled data e.g., terabytes of data from the internet or some other huge data source.

بنحاول نقل Error او الفرق بين المتوقع والحقيقي ونحاول نقل الرقم ده علطول بشكل دوري

- During training, the algorithm performs and evaluates millions of 'fill in the blank' exercises, trying to predict the next element in a sequence e.g., the next word in a sentence, the next element in an image, the next command in a line of code and continually adjusting itself to minimize the difference between its predictions and the actual data.

$$\text{Error} = y - \hat{y}$$

Large Language Models (LLMs)

- In 2017, researchers at Google introduced the transformer architecture, which has been used to develop large language models, like those that power ChatGPT.
- In natural language processing, a transformer encodes each word in a corpus of text as a token and then generates an attention map, which captures each token's relationships with all other tokens.
علاقات بين الكلمات التي موجودة في الجملة <==> بيخليه يفهم سياق الكلام واي التي الكلام يقصده
- This attention map helps the transformer understand context when it generates new text.

Large Language Models (LLMs)

ChatGPT 4o ▾

must to understand
the context of words

apple

Hey! Could you clarify what you mean by "apple"? Are you referring to:

- The fruit?
- The tech company (Apple Inc.)?
- Something else entirely (e.g., a project name or a metaphor)?

Let me know so I can help you better 🍏💻

- A Large Language Model is an AI trained on massive amounts of text data to understand and generate human language.

Large Language Models - Features

Scale | Trained on large datasets (books, articles, websites) with billions of words.

Capabilities | Can answer questions, translate text, summarize, write stories, and more.

Training method | Uses a technique called deep learning—often based on a neural network called a Transformer.

Famous Examples | ChatGPT (by OpenAI), Bard (by Google), Claude (by Anthropic), LLaMA (by Meta).

Large Language Models – How it works

1. Reads a lot of text.
2. Learns patterns (e.g., grammar, facts, logic).
3. Predicts the next word in a sentence based on context.
4. Generates meaningful, coherent text from input prompts.

An LLM doesn't "think" or "understand" like a human—it uses probability and pattern matching to simulate understanding.

Text sentiment

- You can determine the sentiment of a text in Python using the *TextBlob* library, which provides a simple way to analyze text sentiment as "positive," "neutral," or "negative."
- !pip install textblob

```
!pip install textblob
from textblob import TextBlob
```

```
def analyze_sentiment(text):
    # Create a TextBlob object
    blob = TextBlob(text)
    # Get the sentiment polarity (-1 to 1)
    polarity = blob.sentiment.polarity

    # Determine the sentiment
    if polarity > 0:
        return "Positive"
    elif polarity == 0:
        return "Neutral"
    else:
        return "Negative"
```

Test the function

```
text = "I love this product! It's fantastic and works well."
sentiment = analyze_sentiment(text)
print(f"The sentiment of the text is: {sentiment}")
```

فكرة المشروع اعرف هل النص اللي داخلي ده
ايجابي ولا سلبي ولا عادي زي مثلا لما ادخل:

1- "I hate AI" --> Negative, Polarity = less than Zero

1- "I Love AI" --> Positive, Polarity = Greater than Zero

1- "The Weather is Sunny today" --> Neutral, Polarity = Zero

Assignment # 2

- Test the `analyze_sentiment` function on larger dataset of a text file.

يعمل generate new text من الكلام اللي انا كنت كاتبه وكمان يحدد عدد الكلام اللي يكملها بعد
الكلام او الجملة بتاعتي زي :

Text Generation

1- text_generator(prompt, max_length=50)

- Pip3 install transformers
- from transformers import pipeline
- # Load the text generation pipeline
- text_generator = pipeline("text-generation")
- # Provide a prompt
- prompt = "Artificial intelligence is transforming the way we"
- # Generate text
- generated_text = text_generator(prompt, max_length=50)
- # Print the generated text
- print("Generated Text:")
- print(generated_text)

Text Summarization

- Pip3 install transformers
- from transformers import pipeline
- # Load the summarization pipeline
- summarizer = pipeline("summarization")
- # Input text (example: a long paragraph or article)
- text = ""
- Artificial intelligence is a branch of computer science that aims to create systems or machines capable of performing tasks that typically require human intelligence. These tasks include learning, reasoning, problem-solving, understanding language, and even perceiving the environment. In recent years, AI has made significant strides, particularly in areas like natural language processing, computer vision, and robotics.
- ""
- # Generate a summary
- summary = summarizer(text, max_length=60)
- # Print the summary
- print("Summary:")
- print(summary)

هنا اديله نص يلخصهولي

simple python code for chatbot

- `import random`
- `# Predefined responses`
- `responses = {`
- `"hello": ["Hi there!", "Hello!", "Hey! How can I help you?"],`
- `"how are you": ["I'm just a program, but I'm doing fine! How about you?", "I'm good! What about you?"],`
- `"bye": ["Goodbye!", "See you later!", "Take care!"],`
- `"default": ["I'm sorry, I don't understand that. Can you rephrase?"]`
- `}`
- `def chatbot_response(user_input):`
- `user_input = user_input.lower()`
- `for key in responses:`
- `if key in user_input:`
- `return random.choice(responses[key])`
- `return random.choice(responses["default"])`

الشات بوت هيرد عليه بالاجابة response ده بشكل عشوائي
هنا بنعمل شات بوت غرضه ان لو اليوزر كتب اي حاجة بتوافق responses اللي متخزنه لل
اللي موجود فال

simple python code for chatbot

بيقفل المحادثة مع (loop)
وكمان عاملين لو اليوزر كتب كلمة bye الشات بوت (يقفل ال

- # Chat loop

- print("Chatbot: Hi! I'm your chatbot. Type 'bye' to exit.")
- while True:
- user_input = input("You: ")
- if user_input.lower() == "bye":
- print("Chatbot: Goodbye!")
- break
- response = chatbot_response(user_input)
- print("Chatbot:", {response})



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THANK YOU FOR WATCHING

QUESTIONS?

